

CLASSIFICATION of STATES of a MARKOV CHAIN

A state j is:

- ACCESSIBLE** from state i if $p_{ij}^{(n)} > 0$ for some $n \geq 0$
i.e., it can be reached in some steps starting from i .
 - $\Rightarrow$ We say that i and j **COMMUNICATE** if i is accessible from j and j is accessible from i .
 - Any state communicate with itself (this does not mean that there is a self-loop)
 - If i communicate with j and j communicate with k , then i communicate with k (CHAPMAN-KOLMOGOROV)
 - If all states communicate, the Markov chain is called **IRREDUCIBLE**
- TRANSIENT** if the process might never return to this state again, i.e., if there exist a state $i (i \neq j)$ that is accessible from state j but not vice-versa.
 - $\Rightarrow$ there is a positive probability that the process will later move to state i and never return to state j
 - $\Rightarrow$ A transient state is visited only a finite number of times
- RECURRENT** if the process will definitely return to this state again, i.e., it will be visited infinitely often.
 - it is recurrent only if it is not transient
- ABSORBING** if upon entering this state, the process will never leave, i.e., if and only if $p_{jj} = 1$

REMARK: all states in an irreducible finite-state Markov chain are recurrent

- $\Rightarrow$ we can identify an irreducible finite-state Markov chain by showing that all the states communicate
- $\Rightarrow$ A Markov chain is **IRREDUCIBLE** if for any two states i and j there exist $n \in \mathbb{N}$ such that $p_{ij}^{(n)} > 0$

PERIODICITY PROPERTIES

PERIOD of state i : $t \in \mathbb{N}$ st $p_{ii}^{(t)} > 0$ for all n but $t, 2t, 3t, \dots$ and t is the smallest number with this property

$$t = \gcd \{ t \geq 1 : p_{ii}^{(t)} > 0 \}$$

(greatest common divisor)

APERIODIC state: state with period 1 (the process will be in state i at time t and $t+1$ - hence $\gcd = 1$)

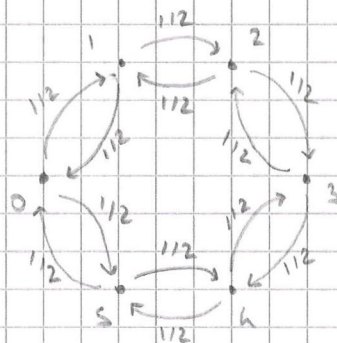
ERGODIC state: recurrent state that is aperiodic

ERGODIC MC: all states are ergodic

EXAMPLE: RANDOM WALK on EVEN-LENGTH cycle

states $\{0, 1, 2, 3, 4, 5\}$

$$p_{ij} = \begin{cases} 1/2 & j = i+1 \pmod{6} \\ 1/2 & j = i-1 \pmod{6} \\ 0 & \text{otherwise} \end{cases}$$



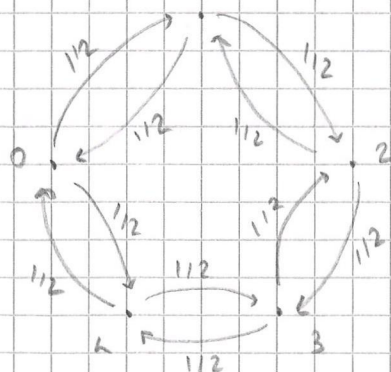
- IRREDUCIBLE
- PERIODIC with PERIOD 2

MODULAR ARITHMETIC
 $a \equiv b \pmod{n}$
 means
 $a = kn + b$

RANDOM WALK on ODD-LENGTH cycle

states $\{0, 1, 2, 3, 4\}$

$$p_{ij} = \begin{cases} 1/2 & j = i+1 \pmod{5} \\ 1/2 & j = i-1 \pmod{5} \\ 0 & \text{otherwise} \end{cases}$$



- IRREDUCIBLE
- APERIODIC

RECURRENT VS ABSORBING

RECURRENT STATE: visited infinitely often

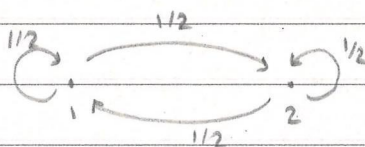
ABSORBING STATE: inescapable

• ABSORBING $\Rightarrow$ RECURRENT

if it is absorbing, it is also recurrent

• The vice-versa is not necessarily true

EX: $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$



ALL THE STATES ARE RECURRENT

NONE OF THEM IS ABSORBING

• Another way to phrase this:

RECURRENT: the process will be back in a recurrent state (with probability 1) at some point, i.e., in any number of steps

ABSORBING: the process will be back in the absorbing state (with probability 1) in one step

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$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = 0$$

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EXERCISE (SUPERVISED)

• Calculate the probs of reaching

- s_2
- s_3
- s_4

Starting at state 0

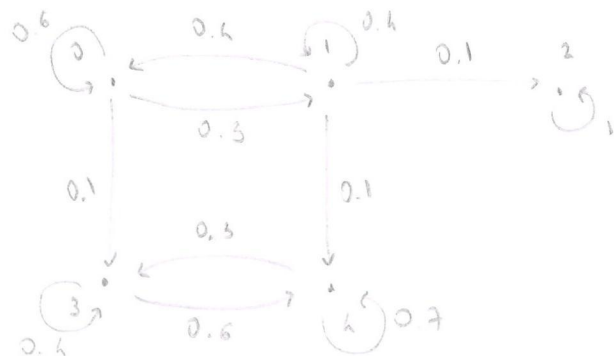
• Calculate $\pi = (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4)$ starting at 0

• On average (long run), how many times is

- s_0

- s_1

visited?



$$P = \begin{bmatrix} 0.6 & 0.3 & 0 & 0.1 & 0 \\ 0.4 & 0.4 & 0.1 & 0 & 0.1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 0.3 & 0.7 \end{bmatrix}$$

• s_3 and s_4 have the same probs because they belong to the same recurrent class

P_x probs of reaching s_2 starting at x

$$\begin{cases} P_0 = 0.6P_0 + 0.3P_1 + 0.1P_3 \\ P_1 = 0.4P_0 + 0.4P_1 + 0.1P_2 + 0.1P_4 \\ P_2 = 1 \\ P_3 = 0 \\ P_4 = 0 \end{cases}$$

$$0.4P_0 = 0.3P_1$$

$$0.6P_1 = 0.4P_0 + 0.1$$

$$P_1 = \frac{2}{3}P_0 + \frac{1}{6}$$

$$P_0 = \frac{3}{4} \left(\frac{2}{3}P_0 + \frac{1}{6} \right) = \frac{1}{2}P_0 + \frac{1}{8}$$

$$P_0 = \frac{1}{4} \quad \text{we go to } s_2 \text{ with probs } \frac{1}{4}$$

$$\Rightarrow 1 - \frac{1}{4} = \frac{3}{4} \text{ probs of going to } \{s_3, s_4\}$$

s_3 and s_4 are transient

$$\Rightarrow \pi_3 = \pi_4 = 0$$

s_2 is absorbing

$$\Rightarrow \pi_2 = \frac{1}{4}$$

s_3 and s_4 are recurrent and we can reach the class with prob $\frac{3}{4}$

$$\Rightarrow \pi_3 + \pi_4 = \frac{3}{4}$$

Now we need to see with what probs we are in s_3 and with what probs we are in s_4 (when not ending in s_2)



$$\pi_{3,4} = (\pi_3, \pi_4)$$

$$P_{3,4} = \begin{bmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{bmatrix}$$

$$\pi_{3,4} P_{3,4} = \pi_{3,4}$$

$$(\pi_3, \pi_4) \begin{bmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{bmatrix} = (\pi_3, \pi_4)$$

$$\pi_3 = 0.4\pi_3 + 0.3\pi_4$$

$$0.6\pi_3 = 0.3\pi_4$$

$$\pi_3 = \frac{1}{2}\pi_4$$

$$\pi_4 = 0.6\pi_3 + 0.7\pi_4$$

$$\frac{1}{2}\pi_4 + \pi_4 = \frac{3}{4}$$

$$\pi_4 = \frac{1}{2}$$

$$\pi_3 = \frac{1}{4}$$

$$\pi = (0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$$

ALTERNATIVE:

$$0.6\pi_3 = 0.3\pi_4$$

$$\pi_{3,4} = \left(\frac{3}{9}, \frac{6}{9} \right) \text{ ratio } 3:6$$

$$= \left(\frac{1}{3}, \frac{2}{3} \right)$$

prob of ending in $\{s_3, s_4\}$

$$\pi = \left(0, 0, \frac{1}{4}, \frac{1}{3} \cdot \frac{3}{4}, \frac{2}{3} \cdot \frac{3}{4} \right)$$

$$= \left(0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2} \right)$$

s_0 and s_1 are transient states, if we reach a recurrent state they will not be visited anymore

Let S_{ij} be the expected number of visits spent in j when starting at i

$$\Rightarrow i \neq j \quad S_{ij} = \sum_k P_{ik} S_{kj}$$

paths from i to k when starting at i

expected number of steps spent at j when starting from k

If k is not a transient state then

$S_{kj} = 0$ (because we cannot go from a recurrent state back to a transient state)

$$- i=j \quad S_{ij} = 1 + \sum_k P_{ik} S_{kj}$$

↓
we are already at the state and we add the number of steps to come back

$$\Rightarrow S_{20} = S_{30} = S_{40} = 0$$

$$S_{21} = S_{31} = S_{41} = 0$$

we have to check $S_{00}, S_{01}, S_{10}, S_{11}$

$\Rightarrow$ we can write the eq in matrix form

$$S = I + P_T S$$

$$\text{where } P_T = \begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.4 \end{bmatrix} \quad \begin{array}{l} \text{Transition sub-matrix} \\ \text{for transient states } s_0 \text{ and } s_1 \end{array}$$

$$S - P_T S = I$$

$$(I - P_T) S = I$$

$$S = (I - P_T)^{-1}$$

$$S = \begin{bmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{bmatrix}$$

$$S = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.4 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 0.4 & -0.3 \\ -0.4 & 0.6 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{\frac{4}{10} \cdot \frac{6}{10} - \frac{3}{10} \cdot \frac{4}{10}} \begin{bmatrix} 6/10 & 3/10 \\ 4/10 & 4/10 \end{bmatrix}$$

$$\frac{2}{5} \cdot \frac{3}{5} - \frac{3}{10} \cdot \frac{2}{5} = \frac{6}{25} - \frac{6}{50} = \frac{12-6}{50}$$

$$= \frac{50}{6} \begin{bmatrix} 6/10 & 3/10 \\ 4/10 & 4/10 \end{bmatrix} = \begin{bmatrix} 5 & 5/2 \\ 10/3 & 10/3 \end{bmatrix}$$

